Analog Circuits vs Digital Circuits

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1 Some case of study

In order to understand how much is the digital and analog part in circuits, we can study some examples of modern chips from big companies. For every chip we can try to undersand how much of it is digital and how much is analog.

1.1 Apple M4

The Apple M4 is a SoC used in some iPads and Macs, it is realized with 3nm technology and obviously is made by a smaller blocks that can be divided betwwen analog, digital or mixed signal, as we can see in fingure 1. [1]

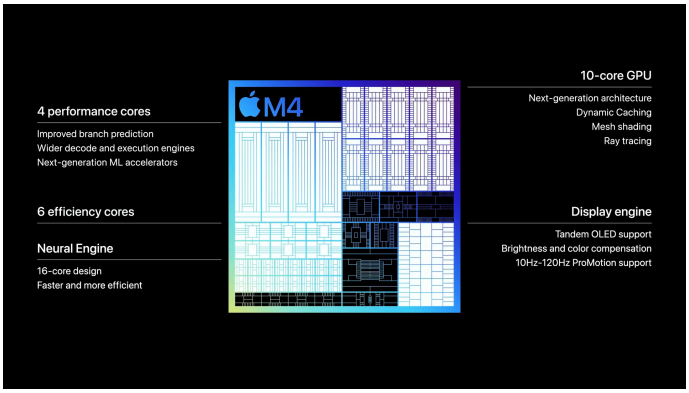


Figure 1: Apple M4 simplified schematic parts

Summuarizing the single blocks and making the separation we can consider the table 1.

Apple M4 Integrated block	Classification
CPU cores	Digital
GPU cores	Digital
Neural Engine	Digital
CPU/GPU caches and SRAM	Digital
Media Engine	Digital
Image Signal Processor (ISP)	Mixed Signal
Display engine	Digital
Memory controllers	Mixed Signal
Thunderbolt / USB controllers	Mixed Signal
Clock generation and PLLs	Analog
Power-management circuits	Analog
High-speed I/O PHYs	Mixed Signal

Table 1: Functional classification of the main blocks of the Apple M4 SoC.

1.2 NVIDIA H100

The NVIDIA H100 is another interesting example, it is a GPU used massivly for AI applications. We can see the simplified schematic in figure 2. [3]



Figure 2: NVIDIA H100 simplified schematic parts

As before we can try to classify all the relevant blocks in order to understand the digital and anlog contibution on the chip. The data is shown in table 2. [2]

NVIDIA H100 Integrated block	Classification
Streaming Multiprocessors (SMs)	Digital
CUDA cores	Digital
Tensor Cores	Digital
L1 cache and shared memory	Digital
L2 cache	Digital
Register files	Digital
GPCs and GPU interconnect	Digital
NVLink controllers	Mixed Signal
HBM memory controllers	Mixed Signal
PCIe interface	Mixed Signal
Clock generation and PLLs	Analog
High-speed I/O PHYs	Mixed Signal
Power-management circuits	Analog

Table 2: Functional classification of the main blocks of the NVIDIA H100 GPU.

1.3 Snapdragon 8 Elite

The Snapdragon 8 Elite is a SoC created by Qualcomm, ised in various smartphones. As the chips before we can divide the schematic in various blocks and classify them, as we can see in table 3. [5] [4]

2 Conclusion

In conlusion, considering the three chips we can say that even if the chips showld be digital-based, still is needed a large amount of analog blcoks, for all three the chips.

References

[1] Apple. Apple introduces m4 chip, 2024.

[2] NVIDIA. Nvidia hopper architecture, 2022.

Snapdragon 8 Elite Integrated block	**Classification**
Oryon CPU cores	Digital
Adreno GPU	Digital
Hexagon NPU	Digital
Shared CPU cache	Digital
AI accelerators	Digital
Sensing Hub	Digital
Spectra image signal processor	Digital
Video processing engine	Digital
Display processing	Digital
Security processor	Digital
5G modem baseband	Digital
5G RF interface	Mixed Signal
Wi-Fi / Bluetooth subsystem	Mixed Signal
UWB interface	Mixed Signal
High-speed memory interface	Mixed Signal
Clock generation and PLLs	Analog
Power-management circuits	Analog

TABLE 3: Functional classification of the main blocks of the Qualcomm Snapdragon 8 Elite SoC.

[3] NVIDIA. Nvidia hopper architecture in-depth, 2022.

[4] Qualcomm Technologies, Inc. Qualcomm unveils snapdragon 8 elite with the world's fastest mobile cpu, 2024.

[5] Qualcomm Technologies, Inc. Snapdragon 8 elite mobile platform product brief, 2024.